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ES26-082 Casa 7, Las Breñas (Yaiza, Lanzarote) - specification in English

Source: *Proyecto Básico de vivienda unifamiliar aislada - Casa nº7* (basic design, not the execution design), architect Cristina González Cabrera (COALZ 3265), Arrecife, July 2023 (88 pages: memoir pp. 3-69, cadastre pp. 70-79, drawings pp. 81-85 at 1:100 on A1, budget p. 87). Enquiry: Pablo Mantello, Kume Arquitectura, 14.10.26 (via Cristian Rodríguez), reply wanted by Wednesday 21.10.26. Client: Chris (private), plot on Calle Las Malvas 2, Las Breñas.

Translated from the memoir sections 1.4.2 (structural system), 2 (constructive memoir) and the drawing legends. Where the basic design says nothing, it is marked **NOT SPECIFIED** and the assumption used in our model is given.

1. The building

- Detached two-storey house, no basement, 240.67 m² built (GF 156.00 m², FF 84.67 m²), 191.21 m² usable; plot 805.85 m², sloping south to north; vehicle access from Calle Las Malvas (south).
- Clear heights 2.80-3.00 m (2.60 m under suspended ceilings).
- Levels (sections A-A / B-B): GF FFL 0.00; FF FFL +3.40; GF roof terrace (solarium) +3.38 with a 0.92 m parapet; main roof +6.60 (structure 0.40); parapet to about +7.00; pool floor -1.52; site levels -0.02 (terraces) down to -2.17 north.
- Programme: GF hall, WC, dressing room, utility, corridor, kitchen-dining-living, bedroom 1, bathroom 1. FF landing, stair, dressing room 2, bathroom 2, WC, bedroom 2, bedroom 3, bathroom 3. Pool, terraces, solarium, cantilever.

2. Foundations (Cimentación)

"The foundation is shallow and consists of reinforced concrete pad footings whose maximum bearing stresses do not exceed the allowable stresses of the ground in any design situation. Centring beams are provided to centre the forces on the footings. Tie beams are provided to prevent relative movement between the foundation elements."

- Ground: medium-density granular soil; allowable bearing **245.3 kN/m²**.
- Founding depth below ground: **-0.80 m (house)** and **-2.80 m (pool / plant room)**.
- Geotechnical study required: construction type C-0, ground group T-1, max 35 m between investigation points, 6 m depth.
- No retaining structures are needed (memoir 1.4.2.2).
- **Our model:** ICF foundation walls (FB) 2 courses, -0.715 to +0.200, on the tie beams; footings / tie beams by others.

3. Structure (Estructura portante)

- Vertical: **reinforced concrete columns** (rectangular) and **steel columns**.
- Horizontal: flat and downstand **RC beams** and **steel beams** carrying the floors.
- No structural drawings in the basic design ("sizes and reinforcement on the project drawings" - not issued).
- Fire: floor, columns, beams R 30.

4. Floors (Forjados) - FLOOR SPEC

*"One-way joist floors (forjados unidireccionales de viguetas): FORJADO 30 - joists at **70 cm centres**, **concrete infill blocks 25 cm**, **5 cm compression topping**, **30 cm total depth**."*

- The sections draw the floor zone 0.40 m (0.30 structure + about 0.10 screed / finishes / ceiling).
- **Our model:** FF floor = 300 mm *forjado* 3,000-3,300 + 100 mm screed and finish 3,300-3,400, over the GF footprint and the FF box (including the 3.0 m cantilever past the GF north face). The GF roof part (solarium) carries the inverted terrace build-up instead of the screed.

5. Ground floor slab (Suelos en contacto con el terreno) - SLAB SPEC

"Floors in contact with the ground are a concrete ground-bearing slab (solera de hormigón), whose thicknesses and layers will be reviewed before the execution design is prepared. Any ground improvement works needed will be included."

- **NOT SPECIFIED** (thickness, sub-base, membrane).
- **Our model (indicative):** 150 mm concrete slab -150..0 on a 150 mm compacted sub-base -300..-150, inside the ICF walls. Pool base slab 300 mm under -1.52 (assumed); plant room floor slab 300 mm (assumed).

6. Roofs (Cubiertas y terrazas) - ROOF SPEC

Main roof (non-walkable flat roof, +6.60):

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*"Roof: falls in **lightweight concrete, 10 cm average thickness**; **modified oxidised-bitumen membrane with polyethylene reinforcement**; **rigid extruded polystyrene (XPS) insulation board**; separating layer and **6 cm gravel ballast** as protection."*

Terraces (GF roof solarium +3.38, accessible inverted flat roof):

*"Accessible flat inverted terrace: **lightweight concrete 10 cm average** laid to falls, **1:6 mortar 2 cm**, **bitumen membrane** floating except at the perimeter and singular points, protective layer, finished with **insulating paving tiles (XPS base + fibre-reinforced polymer mortar), laid dry**."*

- Structural deck under both: the forjado 30 (section 4) - the sections draw 0.40.
- Gutter / outlet details on sheet 04: LIMAHOYA (valley), QAN-12, QAN-13, QAT-16, QAT-15 (drawn details, no text spec).
- **Our model:** main roof = forjado 300 at 6,200-6,500 + build-up 6,500-6,600 inside the FF box, parapet walls to +7.00; solarium = forjado at 3,000-3,300 + terrace build-up 3,300-3,380 inside the 0.92 m parapet (+4.30).

7. Façades and partitions as designed (Fachadas y particiones) - what the ICF replaces

*"Double-leaf façade, **U = 0.558 W/m²K** (excluding integrated thermal bridges): outer leaf **12 cm hollow vibrated concrete block** (R 0.23 m²K/W), inner leaf **9 cm hollow concrete block** (R 0.19), laid in M-10 mortar; cavity insulation **3 cm XPS board, λ 0.034 W/mK**; external render GP CS IV W1 1.5 cm (λ 0.67) finished GP CS III W1 0.5 cm (λ 0.47); internal gypsum 1.5 cm (λ 0.30)."*

- At singular points the façades are clad in **local volcanic stone** (detail to the execution design). Drawing legend: 1 render painted white; 2 anodised aluminium joinery, dark wood colour; 3 volcanic stone cladding 20 cm; 4 timber joinery, dark wood colour; 5 perimeter wall rendered and painted white, max height 2.00 m; 6 volcanic stone wall 40 cm.
- Partitions: 9 or 12 cm vibrated concrete block, projected gypsum both faces.
- Drawn wall thickness: GF 0.40 m, FF 0.30 m.

8. Thermal (DB-HE 1)

- Climate zone **A3** (Yaiza). DB-HE 1 wall limit (Pablo) **U ≤ 0.70 W/m²K**; the designed block wall is 0.558.
- Basic design asks for no special insulation or lambda.
- Nudura 6in XR35 (4in EPS each face) and 6in Standard (2⅝in EPS each face) both beat 0.70 by a wide margin; **the U-value of the configuration and the declared λ of the Nudura EPS must come from the Nudura technical data sheet** (requested for the HE1 justification - to follow with the quotation).

9. Pool (Piscina) and water tank (Aljibe)

- Drawn pool 8.00 x 2.50 m (20.00 m²), floor -1.52; **client extends to 10,000 x 3,600 mm, 1.5 m deep** (Pablo 14.10), with **pump / storage room behind** (client sketch: about 2,000 mm deep, north side).
- Water tank (aljibe) **6,470 litres** (memoir 1.3, regulation of the island water authority) - location not drawn; Pablo is asking whether the plant room and aljibe are ICF too. **Not modelled.**

10. Joinery schedule (Memoria de carpintería, sheet 04)

Ref	W x H (m)	Type	Where (our reading)
P1	1.68 x 3.00	timber entrance door (red)	GF west, hall
P2	10.85 x 3.00	anodised aluminium sliding wall (3.21 + 4.44 + 3.21)	GF north, living
P3	3.00 x 3.00	aluminium sliding	GF east, bedroom 1 to solana
P4	3.33 x 2.80	aluminium sliding	FF west, bedroom 3 to solarium
P5	1.00 x 3.00	aluminium door	FF west, bathroom 2 (modelled 2.80)
P6	6.16 x 2.80	aluminium fixed + tilt-turn	FF north, bedroom 2 - the full wall
P7 (5), P8, P9 (2)	0.80 / 1.00 / 1.20	lacquered timber internal doors	internal - not ICF
P10	4.56 x 1.40	corten steel and timber gate	site - not ICF
V1 / V2 / V3	0.90 / 1.20 / 0.60 x 2.00, sill 1.00	aluminium tilt-turn	GF south / west
V4 / V5	0.50 / 0.93 x 1.90, sill 0.90	aluminium tilt-turn	FF south / east

11. ICF scope and spec (our determination)

The client's green highlights (IMG_9040.jpeg) are the ICF walls:

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- **GF:** the full perimeter of the 13.19 x 10.98 m box and the full-width cross wall between the living space and the bedroom wing. **FF:** both side walls of the 6.76 x 14.28 m box (full length to the cantilever slab edge), the south wall, and the stair wall. The north "wall" of bedroom 2 is the P6 glazing, floor to ceiling - no ICF there. **Pool:** the pool box and the pump / storage room behind it.
- **External walls: Nudura 6in XR35** (BuildEase standard external wall).
- **Internal walls (GF cross wall, FF stair wall): 6in Standard** - no thermal duty.
- **Pool and plant room: 6in Standard** (as ES26-078 Busot) - pool walls 4 courses, plant room 6 courses (founded -2.80).
- **Solarium parapet:** 0.92 m over the GF roof terrace, 6in XR35, cut flat at +4.30.
- The frame is RC + steel: the ICF is the envelope and its core size is for the engineer. Modelled as drawn, including the FF box cantilevering 0.30 past the GF south face and 3.0 m past the GF north face.

12. Budget (Resumen de presupuesto, p. 87) - for reference

Material execution total **EUR 288,050.94** (+13% overheads). Chapters: earthworks 3,633.59; foundations 26,154.66; structure 31,026.88; masonry and renders 47,101.51; waterproofing and insulation 5,409.69; floor / wall finishes 51,070.52; sanitaryware 5,560.38; ceilings 5,089.00; timber joinery 8,541.75; aluminium joinery and metalwork 34,984.99; roofs 6,131.11; painting 8,729.75; electrics 7,345.44; telecoms 2,994.16; plumbing and drainage 12,773.75; ventilation 1,740.85; pool 6,129.31; sundries 610.12; landscaping 6,714.33; health and safety 8,159.69; quality control 3,186.21; waste 4,963.25.

13. Open questions for Kume

- Pool position after the extension (we kept the drawn west edge and the 2.50 m to the house) and the plant room depth / height.
- Is the aljibe (6,470 l) ICF, and where is it?
- GF slab build-up (not specified) and the pool / plant room floor and roof slabs.
- Structural design (column / beam positions, core size) - not in the basic design.
- Delivery address, building licence, builder, wall start date (Pablo is chasing).